

SAPIENZA
UNIVERSITÀ DI ROMA

Programmazione lineare intera

Catalogo dei problemi MIP

I problemi di programmazione lineare intera e mista usati negli esempi e negli esercizi di tutto il modulo, citati con la loro sigla ($P0$, $P1$, …): problemi con due, tre e più variabili, divisi per tipo. Per ciascuno il problema e la figura della regione ammissibile.

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Versione online e materiale scaricabile:
fabiofurini.github.io/programmazione-lineare-intera/
github.com/fabiofurini/programmazione-lineare-intera

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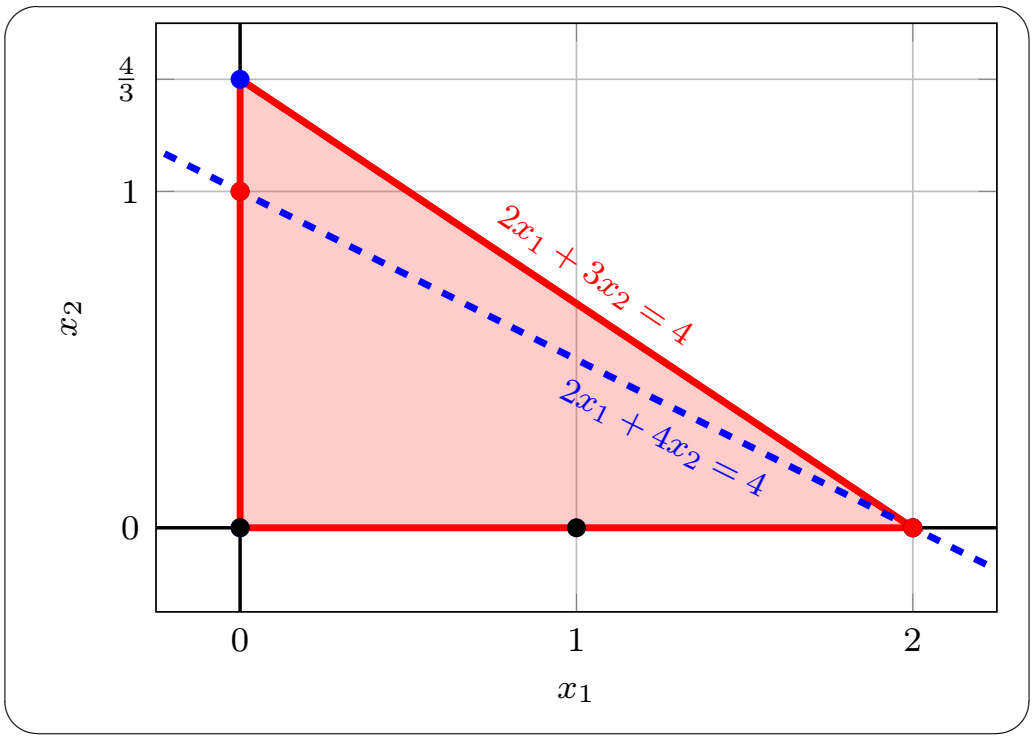
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Capitolo 1

Problemi di PLI con due variabili – Tipo A

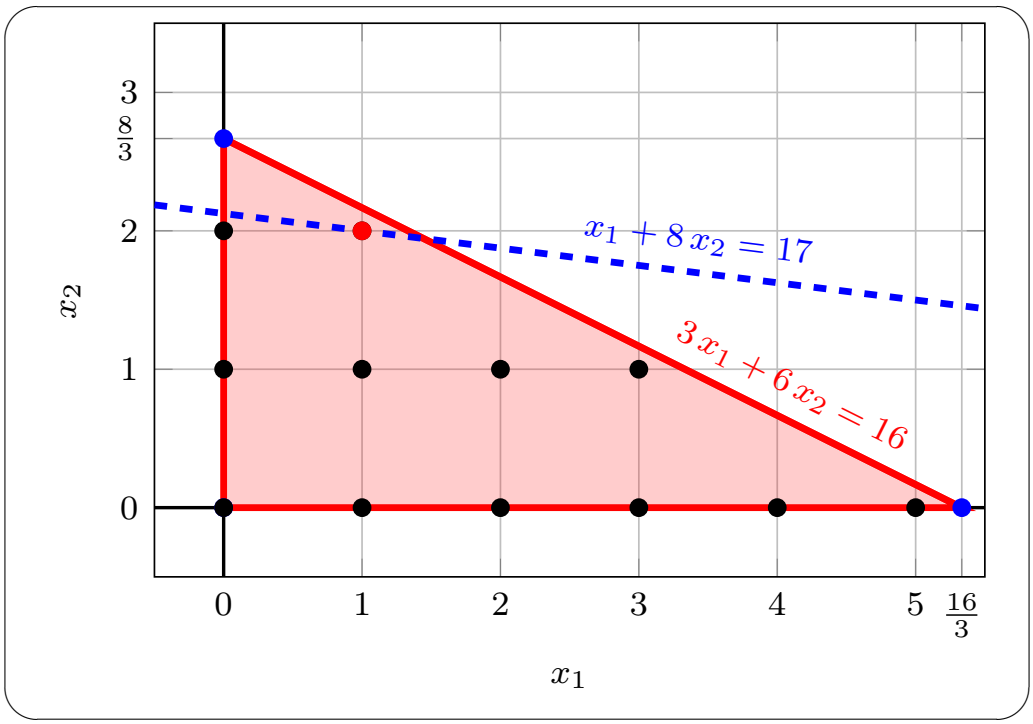
1.1 Problema di PLI *P0*

$$
\begin{array}{lrcrcrl}
\max & 2 & x_1 & + & 4 & x_2 & \\
\text{soggetto a} & 2 & x_1 & + & 3 & x_2 & \leq \quad 4 \\
 & & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$



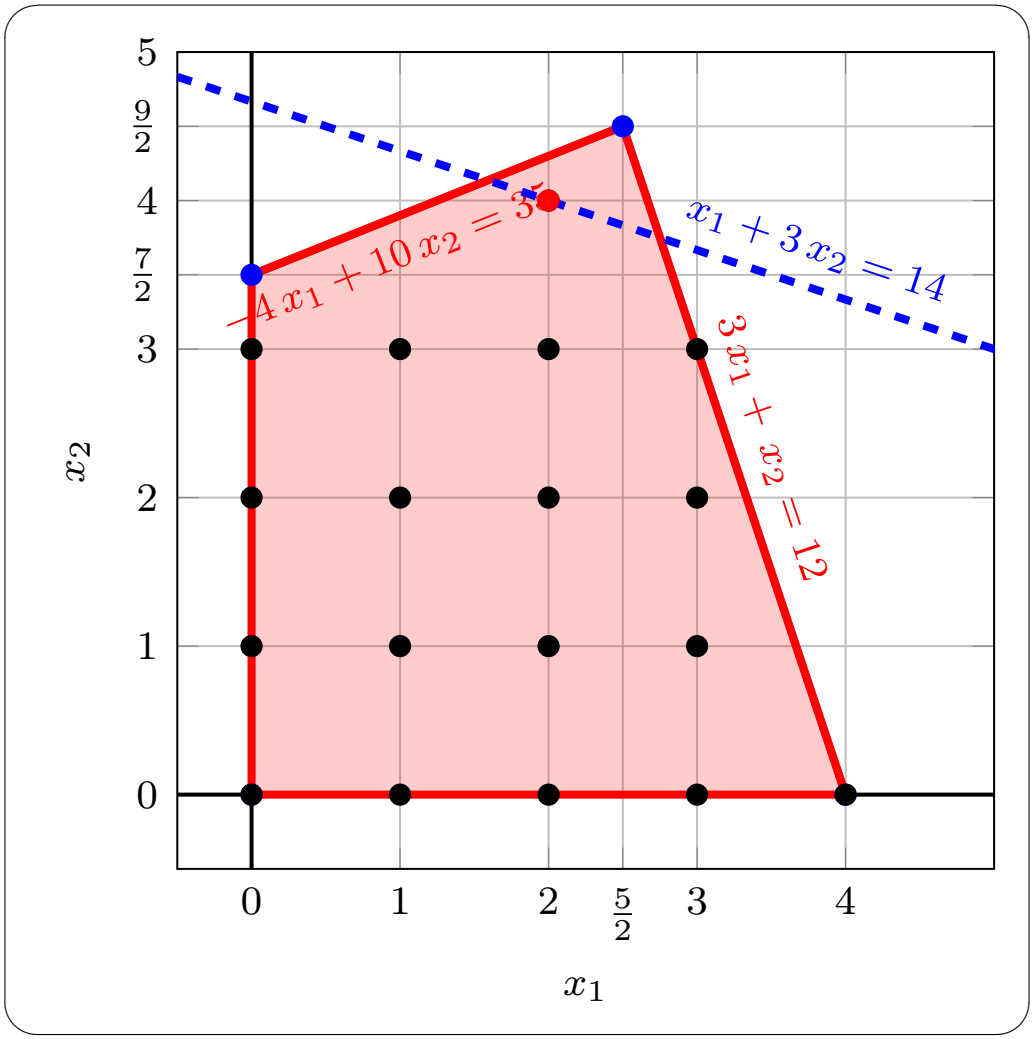
1.2 Problema di PLI $P1$

$$
\begin{array}{lrcrcrl}
\max & & x_1 & + & 8\ x_2 & & \\
\text{soggetto a} & 3\ x_1 & + & 6\ x_2 & \leq & 16 & \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



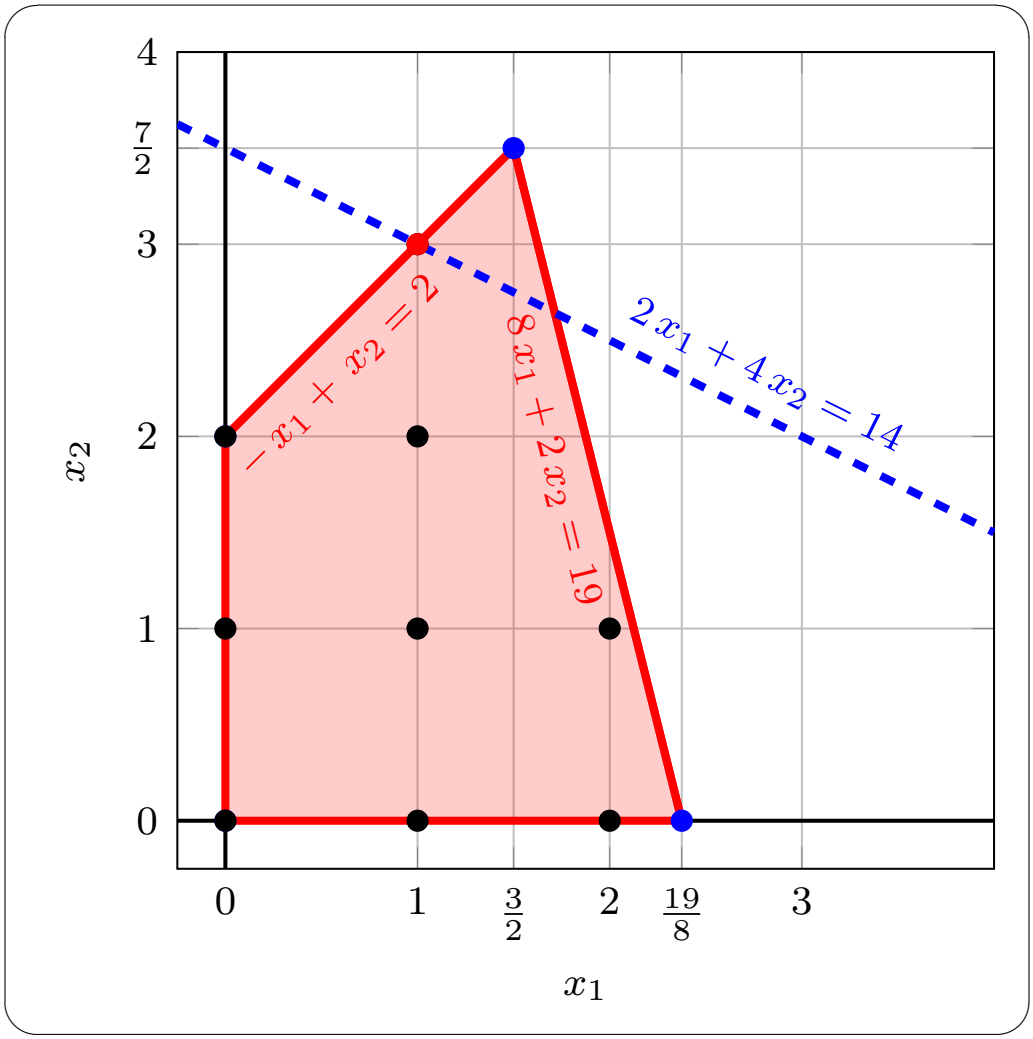
1.3 Problema di PLI *P2*

$$
\begin{array}{lrcrcrcl}
\max & & x_1 & + & 3 & x_2 & & \\
\text{soggetto a} & -4 & x_1 & + & 10 & x_2 & \leq & 35 \\
& 3 & x_1 & + & & x_2 & \leq & 12 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



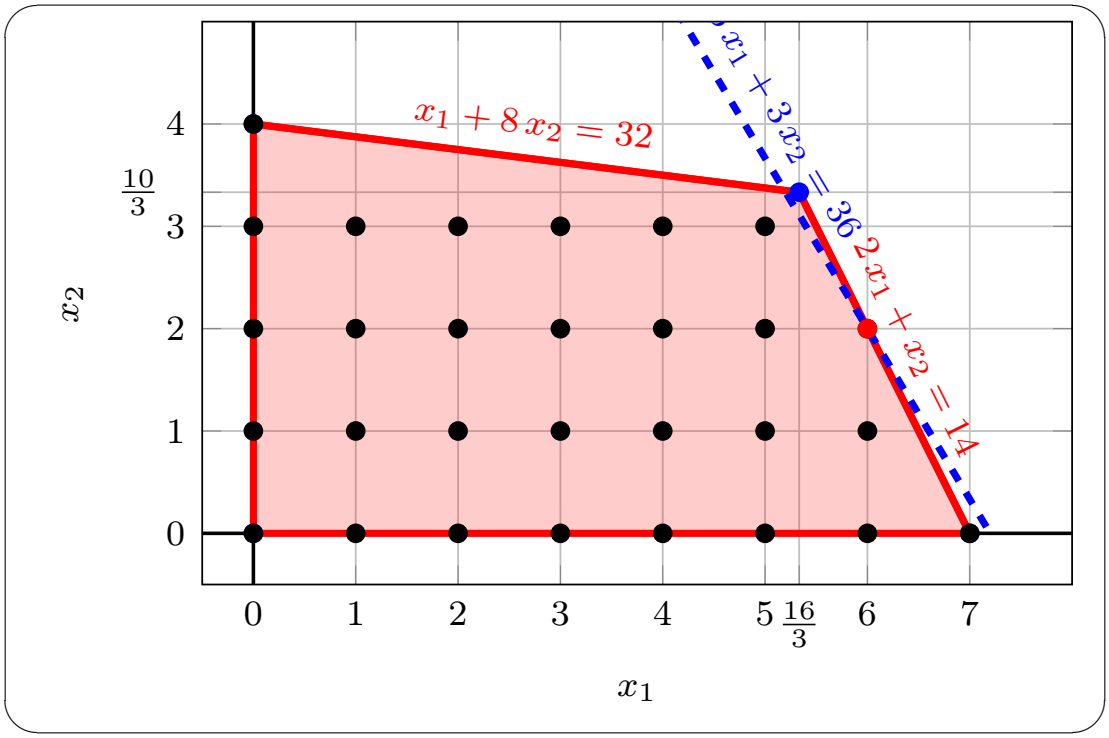
1.4 Problema di PLI *P3*

$$
\begin{array}{rrcrcrl}
\max & 2 & x_1 & + & 4 & x_2 & \\
\text{soggetto a} & -1 & x_1 & + & 1 & x_2 & \leq 2 \\
& 8 & x_1 & + & 2 & x_2 & \leq 19 \\
& & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$



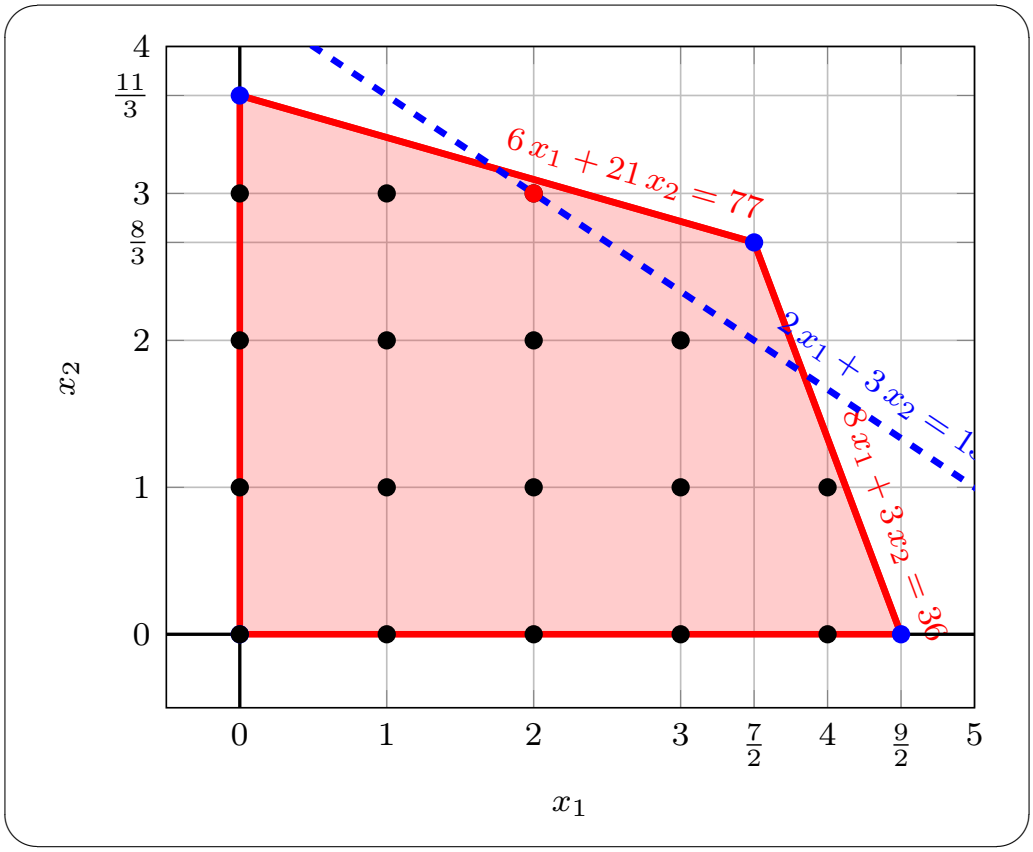
1.5 Problema di PLI *P4*

$$
\begin{array}{lrcrcrl}
\max & 5\,x_1 & + & 3\,x_2 & & \\
\text{soggetto a} & 2\,x_1 & + & x_2 & \leq & 14 \\
& x_1 & + & 8\,x_2 & \leq & 32 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



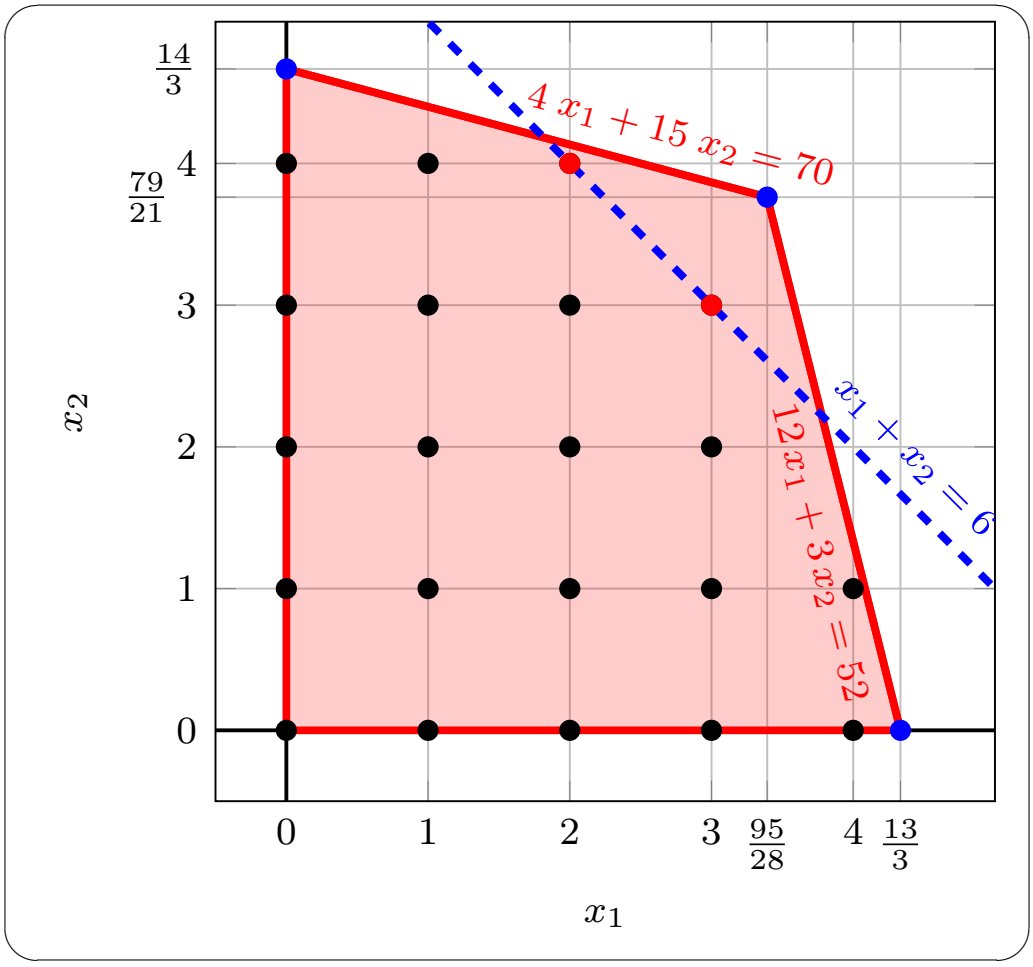
1.6 Problema di PLI $P5$

$$
\begin{array}{rrcrcrl}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{soggetto a} & 6\,x_1 & + & 21\,x_2 & \leq & 77 \\
& 8\,x_1 & + & 3\,x_2 & \leq & 36 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



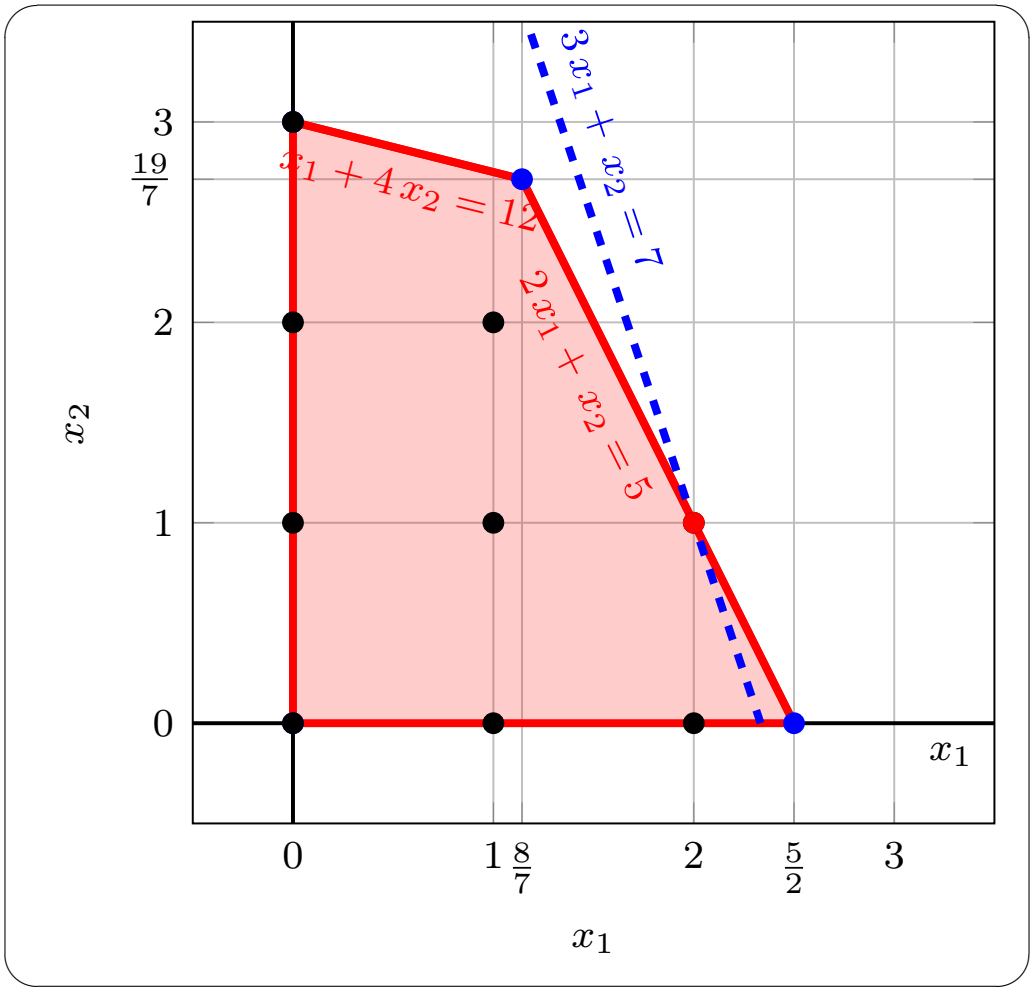
1.7 Problema di PLI *P6*

$$
\begin{array}{rrcrcrr}
\max & & x_1 & + & & x_2 & \\
\text{soggetto a} & 4 & x_1 & + & 15 & x_2 & \leq & 70 \\
& 12 & x_1 & + & 3 & x_2 & \leq & 52 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



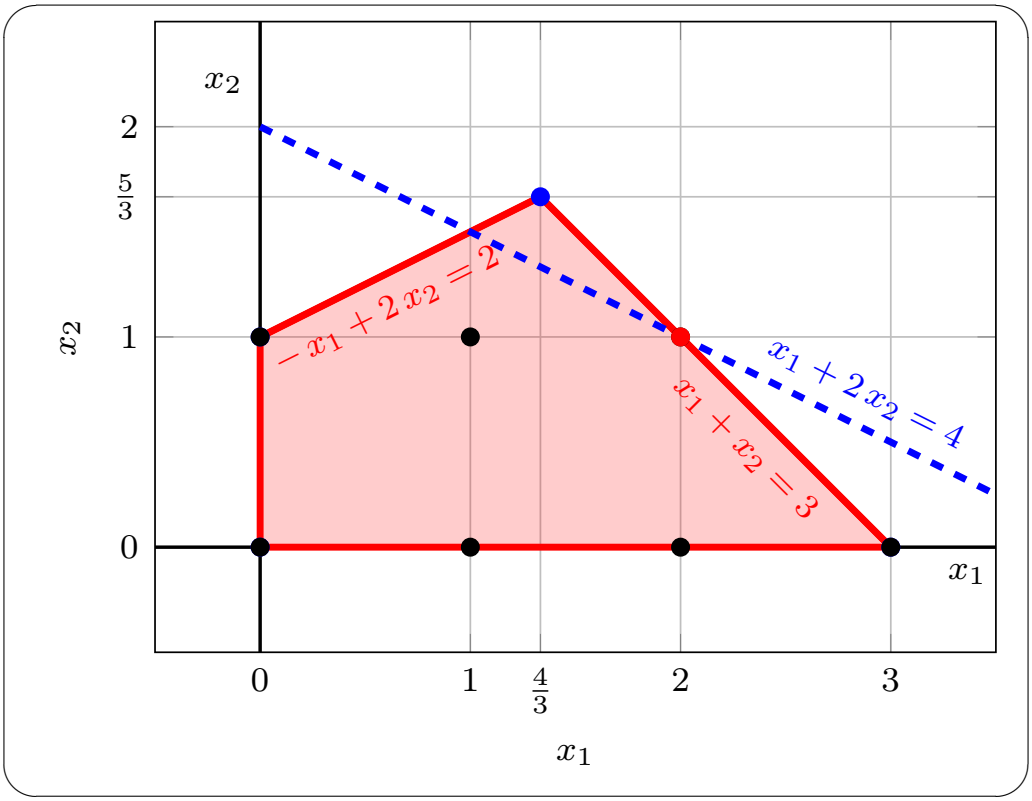
1.8 Problema di PLI *P7*

$$
\begin{array}{rrrrrrrr}
\max & 3 & x_1 & + & & x_2 & & \\
\text{soggetto a} & & x_1 & + & 4 & x_2 & \leq & 12 \\
& 2 & x_1 & + & & x_2 & \leq & 5 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



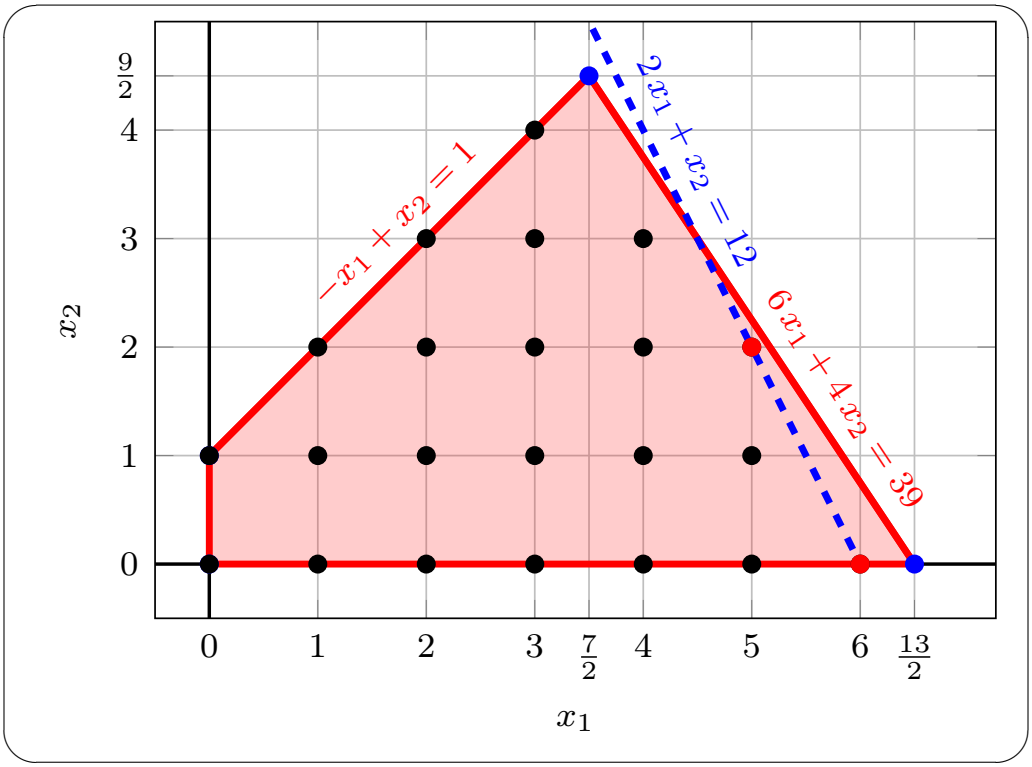
1.9 Problema di PLI *P8*

$$
\begin{array}{lrcrcl}
\max & x_1 & + & 2\ x_2 & & \\
\text{soggetto a} & -\ x_1 & + & 2\ x_2 & \leq & 2 \\
 & x_1 & + & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



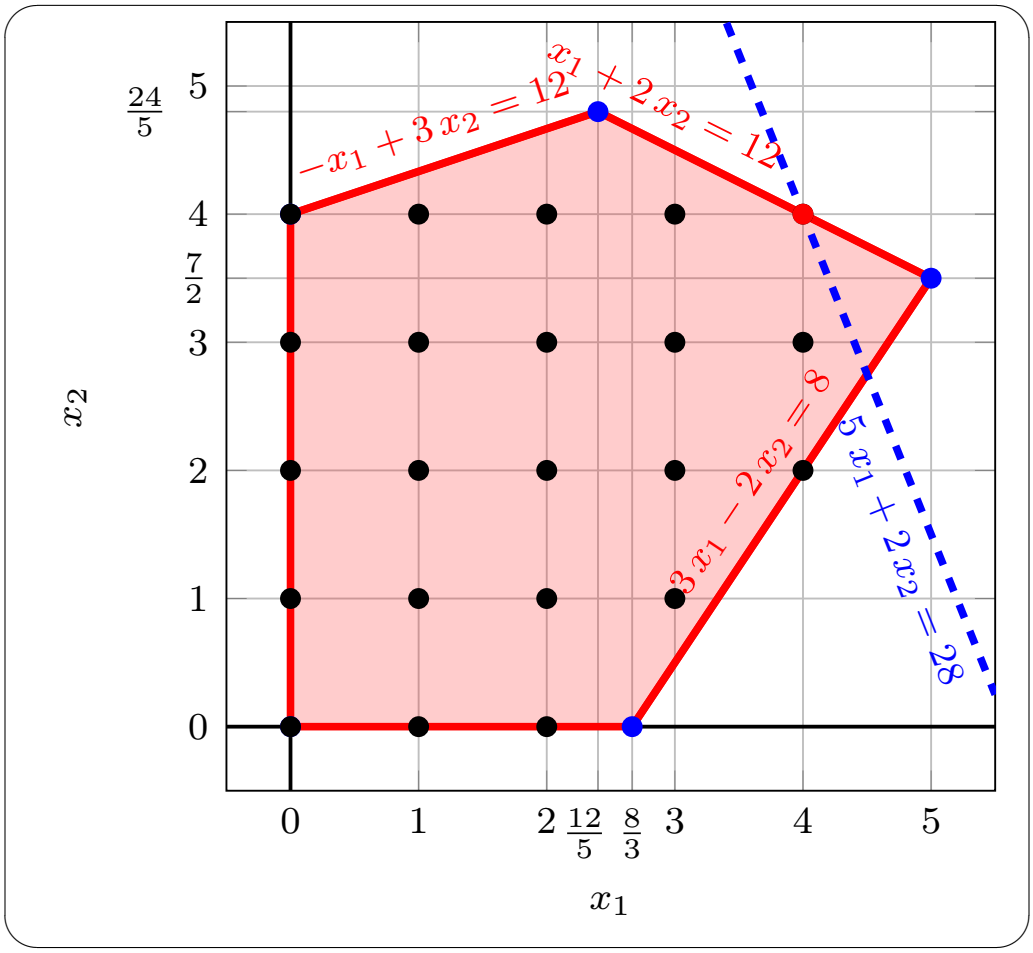
1.10 Problema di PLI *P9*

$$
\begin{array}{rrrrrrrr}
\max & 2 & x_1 & + & & x_2 & & \\
\text{soggetto a} & - & x_1 & + & & x_2 & \leq & 1 \\
& 6 & x_1 & + & 4 & x_2 & \leq & 39 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



1.11 Problema di PLI *P10*

$$
\begin{array}{rrcrcrl}
\max & 5 & x_1 & + & 2 & x_2 & \\
\text{soggetto a} & - & x_1 & + & 3 & x_2 & \leq \quad 12 \\
 & & x_1 & + & 2 & x_2 & \leq \quad 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq \quad 8 \\
 & & x_1, & & & x_2 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$

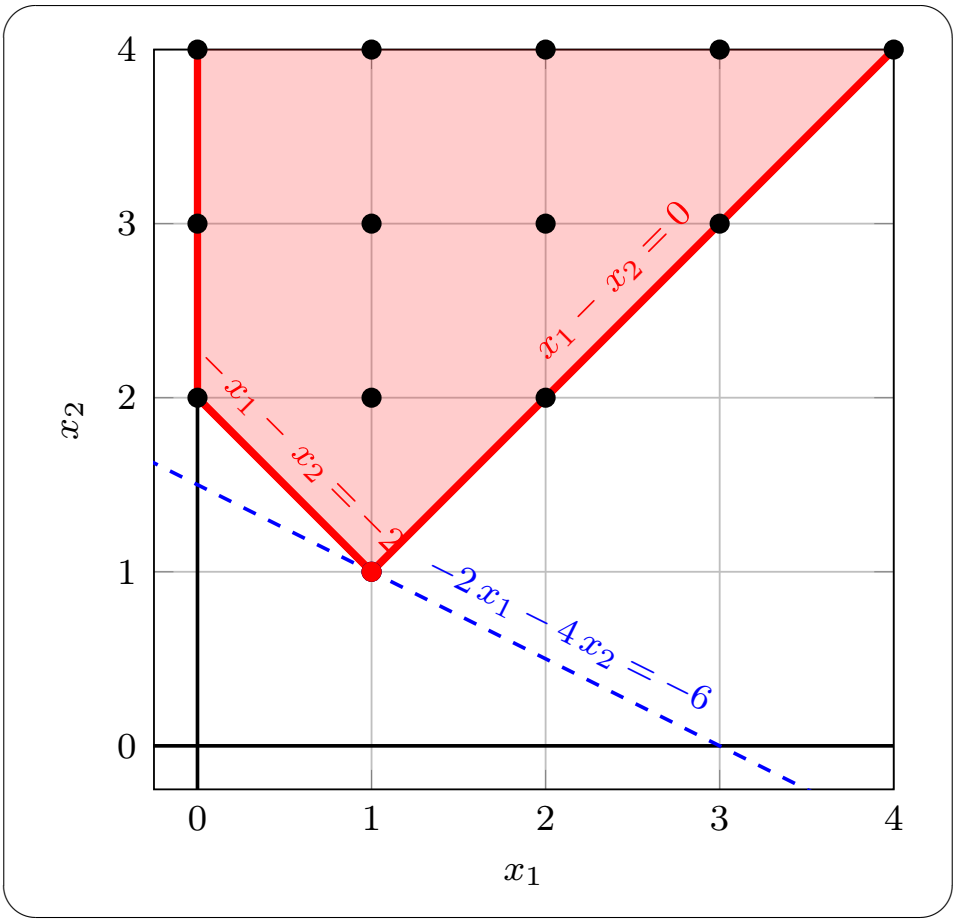


Capitolo 2

Problemi di PLI con due variabili – Tipo B

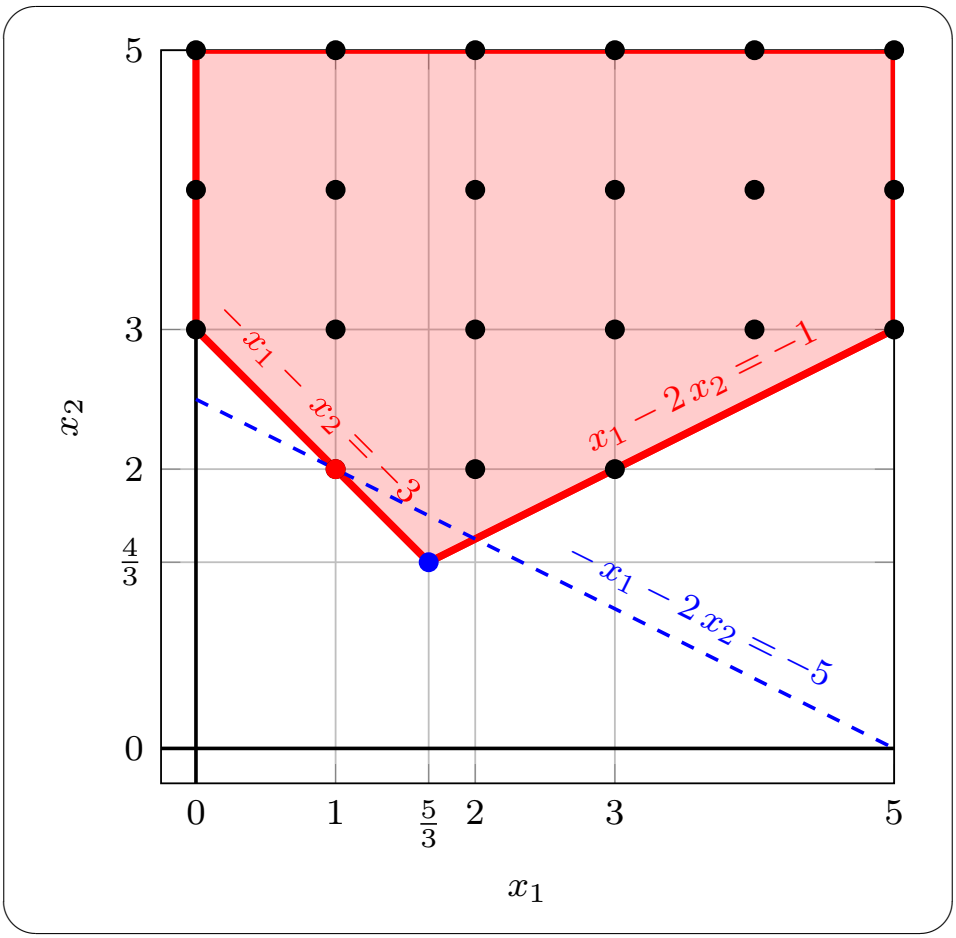
2.1 Problema di PLI *P11*

$$
\begin{array}{lrcrcl}
\max & -2\,x_1 & - & 4\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -2 \\
 & x_1 & - & x_2 & \leq & 0 \\
 & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



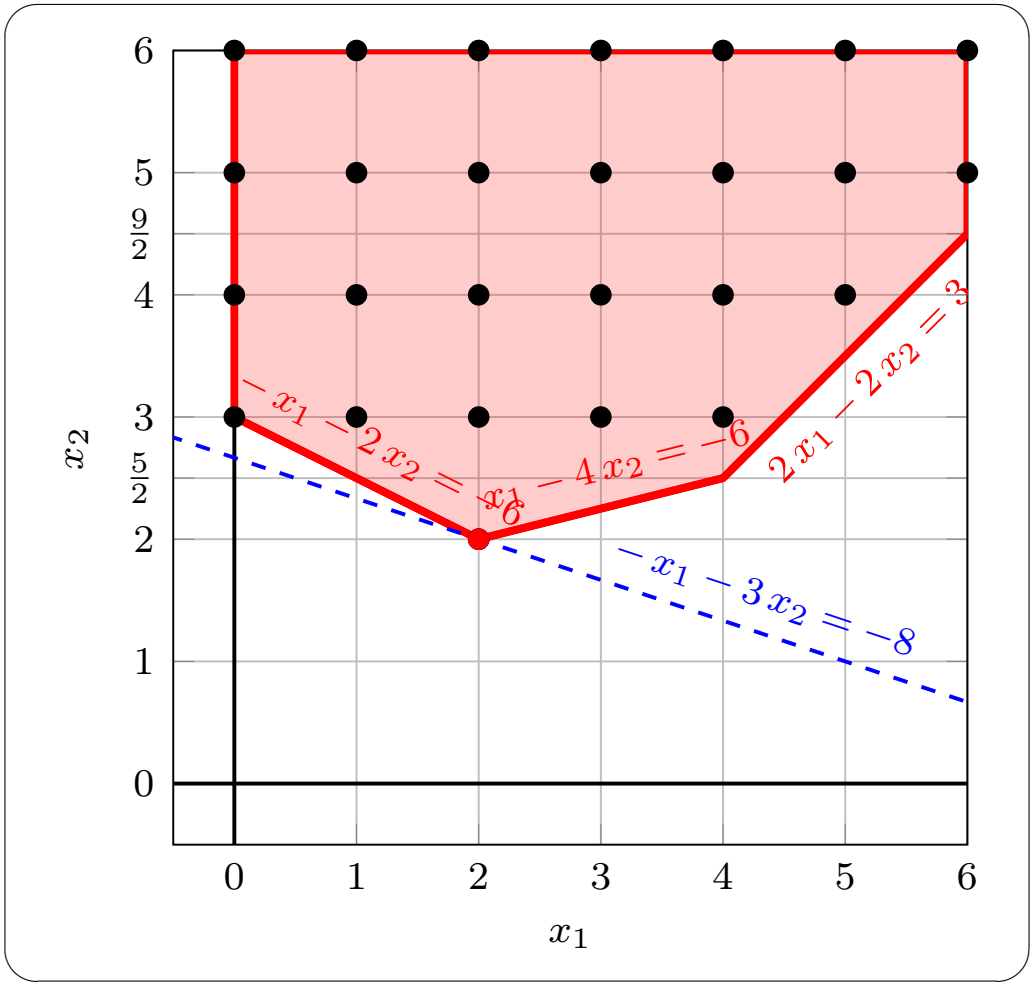
2.2 Problema di PLI *P*12

$$
\begin{array}{rrrrrrr}
\max & -x_1 & - & 2\,x_2 & & \\
\text{soggetto a} & -x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & -1 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



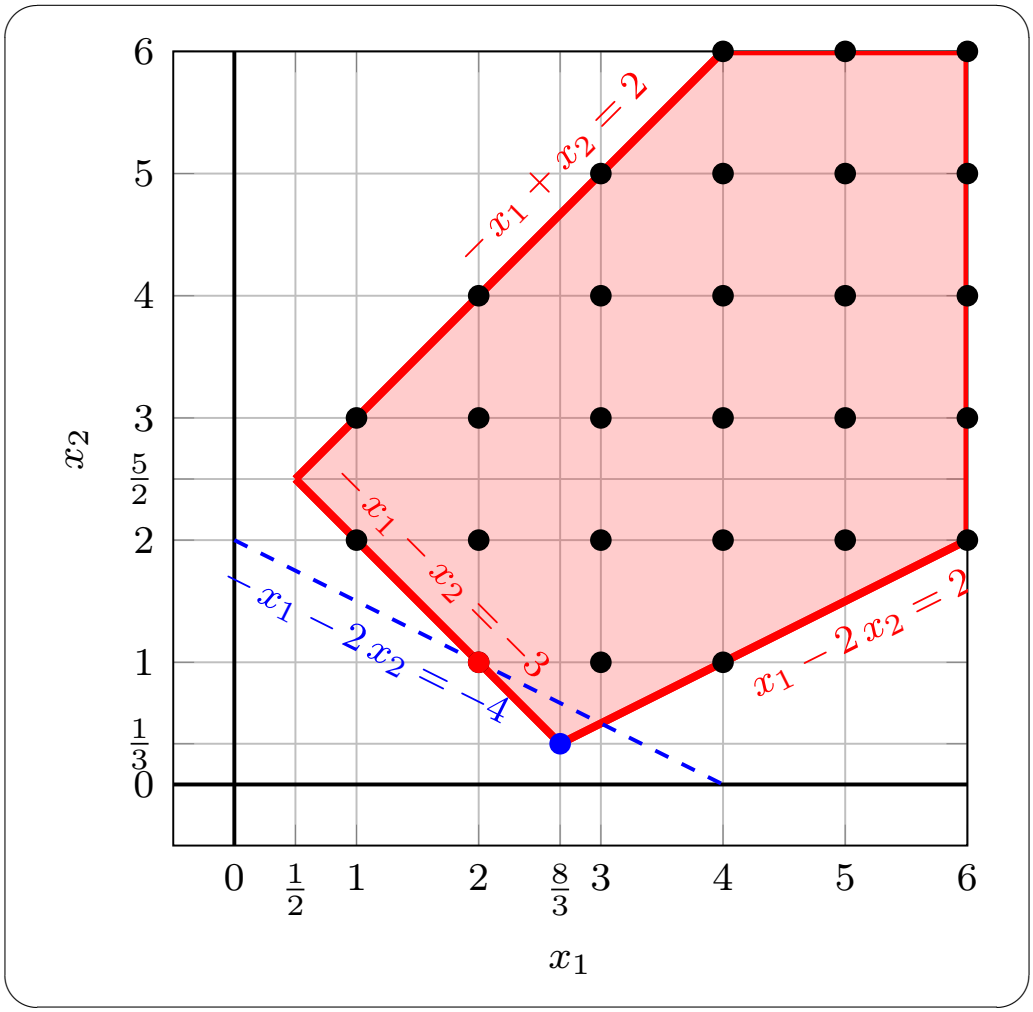
2.3 Problema di PLI $P13$

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & \\
\text{soggetto a} & - & x_1 & - & 2 & x_2 & \leq & -6 \\
 & & x_1 & - & 4 & x_2 & \leq & -6 \\
 & 2 & x_1 & - & 2 & x_2 & \leq & 3 \\
 & & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



2.4 Problema di PLI *P*14

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 2 & x_2 & & \\
\text{soggetto a} & - & x_1 & + & & x_2 & \leq & 2 \\
& & x_1 & - & 2 & x_2 & \leq & 2 \\
& - & x_1 & - & & x_2 & \leq & -3 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

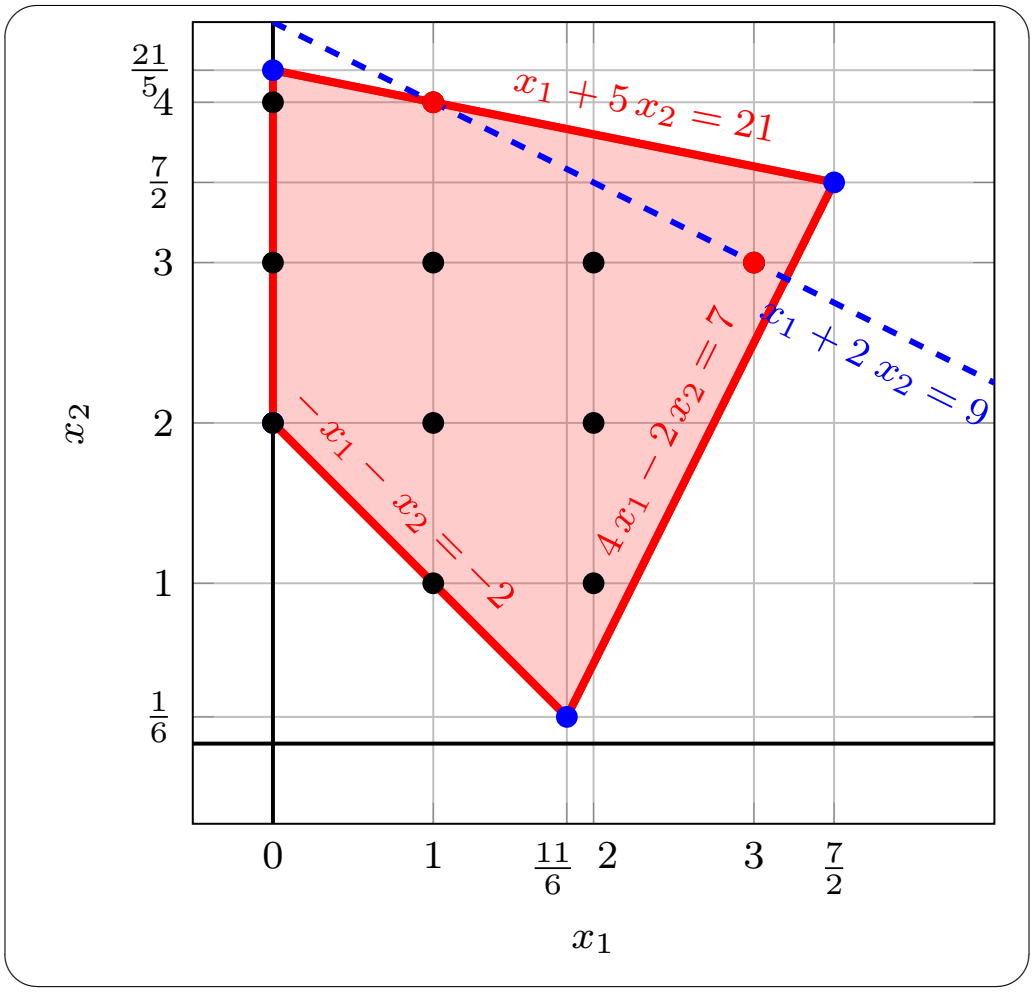


Capitolo 3

Problemi di PLI con due variabili – Tipo C

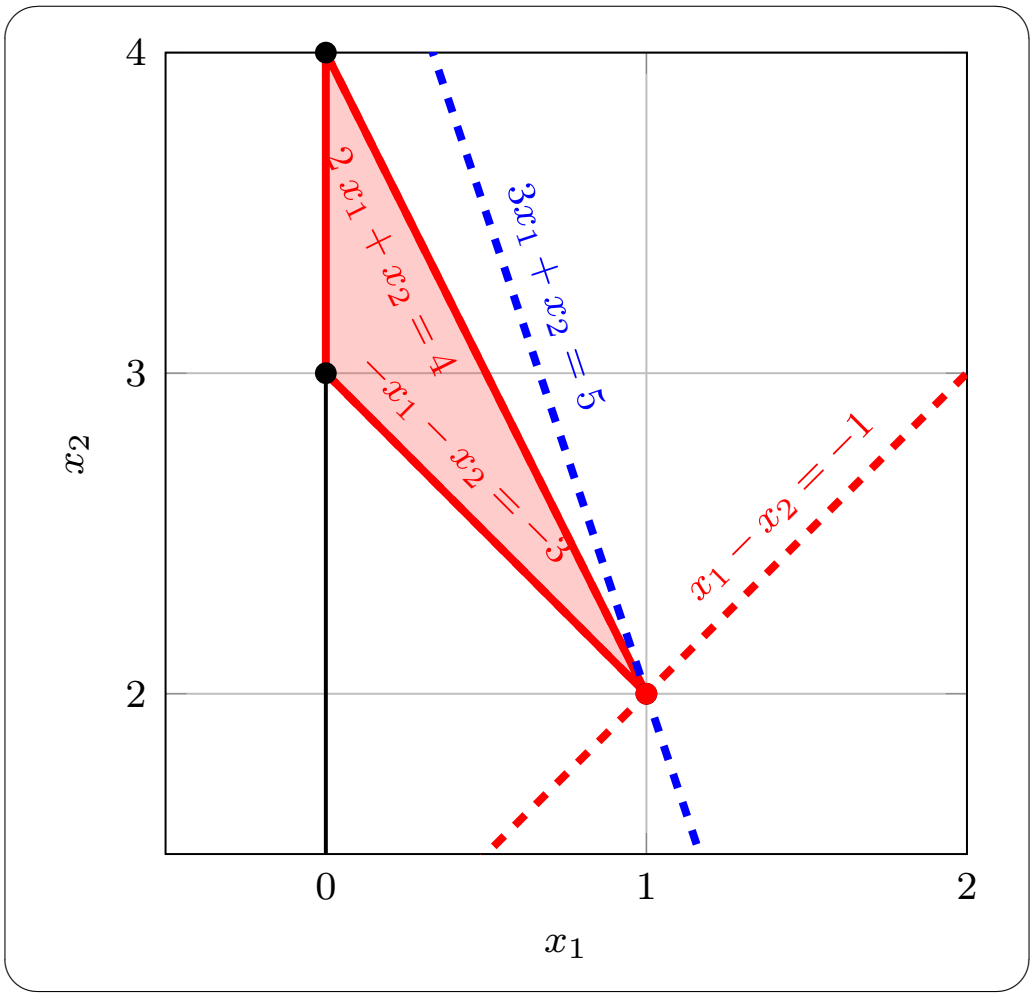
3.1 Problema di PLI *P*21

$$
\begin{array}{rrrrrrrr}
\max & x_1 & + & 2 & x_2 & & \\
\text{soggetto a} & -x_1 & - & & x_2 & \leq & -2 \\
& 4\,x_1 & - & 2 & x_2 & \leq & 7 \\
& x_1 & + & 5 & x_2 & \leq & 21 \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



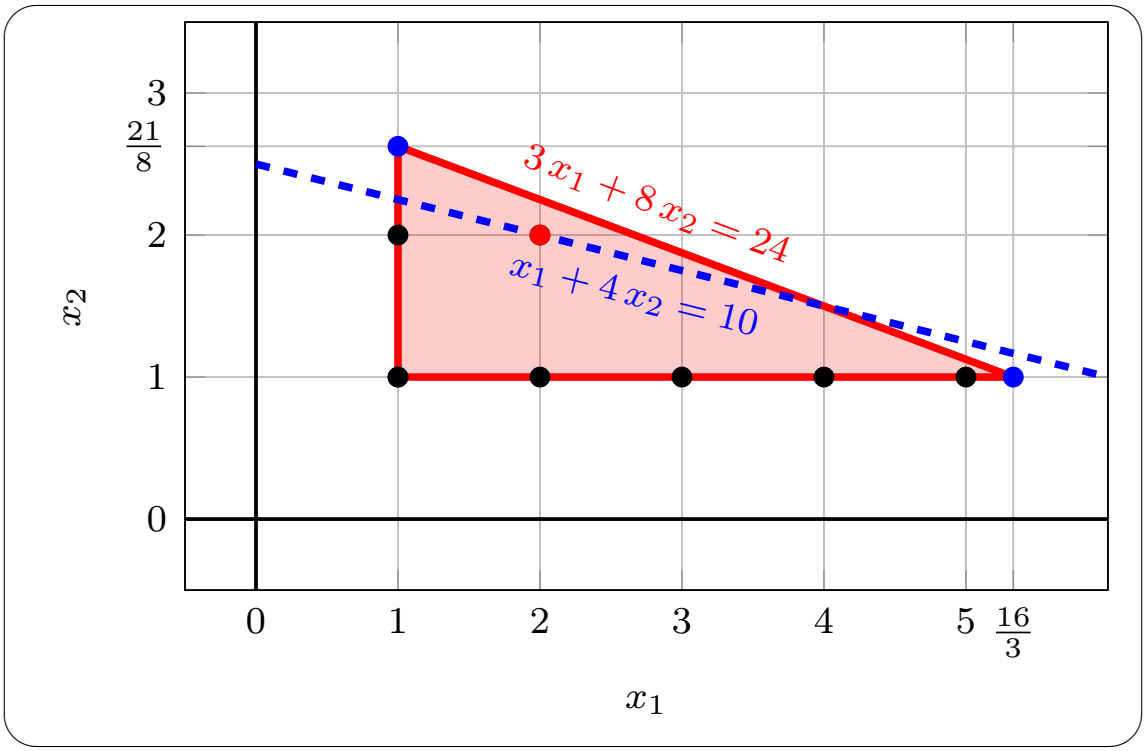
3.2 Problema di PLI *P22*

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & - & x_2 & \leq & -1 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



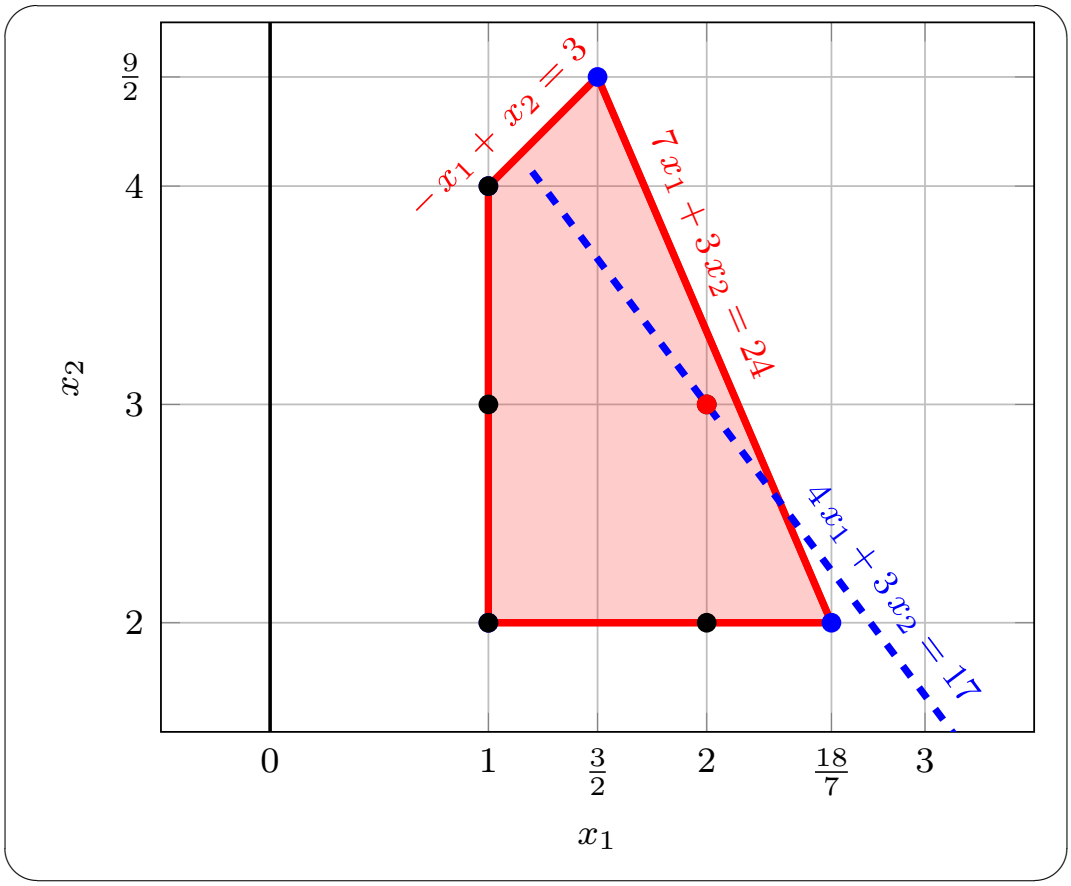
3.3 Problema di PLI *P*23

$$
\begin{array}{lrcrcrr}
\max & & x_1 & + & 4 & x_2 & \\
\text{soggetto a} & 3 & x_1 & + & 8 & x_2 & \leq & 24 \\
 & & x_1 & & & & \geq & 1 \\
 & & & & & x_2 & \geq & 1 \\
 & & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



3.4 Problema di PLI *P24*

$$
\begin{array}{rrcrcrcr}
\max & 4 & x_1 & + & 3 & x_2 & & \\
\text{soggetto a} & 7 & x_1 & + & 3 & x_2 & \leq & 24 \\
& - & x_1 & + & & x_2 & \leq & 3 \\
& & x_1 & & & & \geq & 1 \\
& & & & & x_2 & \geq & 2 \\
& & x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

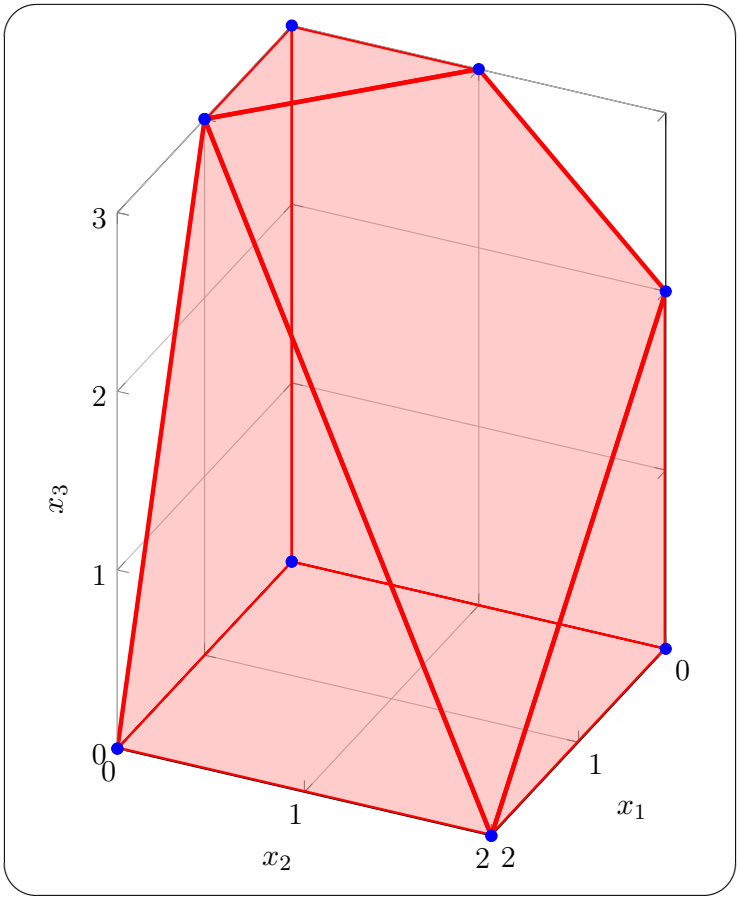


Capitolo 4

Problemi di PLI con tre variabili – Tipo A

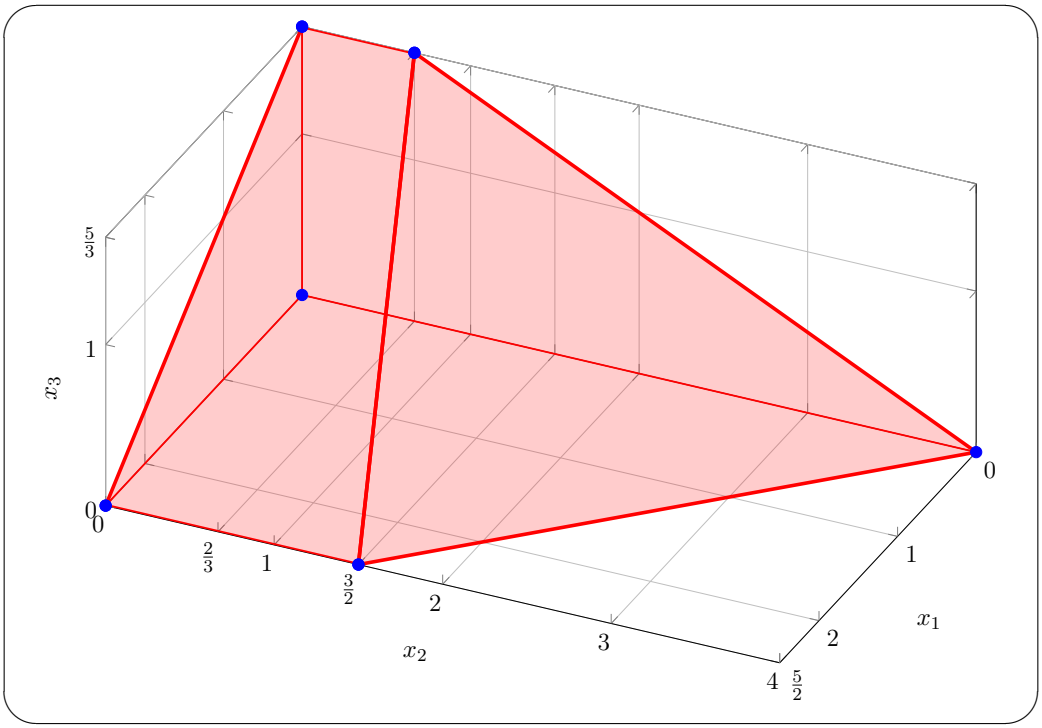
4.1 Problema di PLI *P41*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & x_2 & + & 3\,x_3 & & \\
\text{soggetto a} & x_1 & + & x_2 & + & x_3 & \leq & 4 \\
 & & & x_2 & & & \leq & 2 \\
 & & & & & x_3 & \leq & 3 \\
 & 3\,x_1 & & & + & x_3 & \leq & 6 \\
 & x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



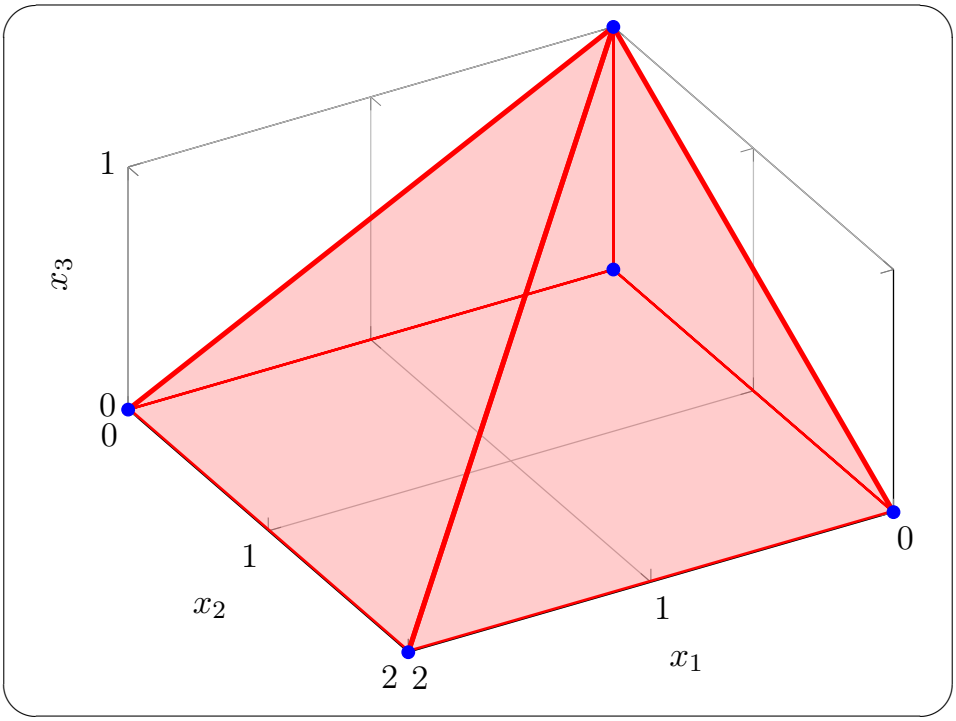
4.2 Problema di PLI *P42*

$$
\begin{array}{rrcrcrcrcl}
\max & 3 & x_1 & + & 2 & x_2 & + & 4 & x_3 & \\
\text{soggetto a} & & x_1 & + & & x_2 & + & 2 & x_3 & \leq & 4 \\
& 2 & x_1 & & & & + & 3 & x_3 & \leq & 5 \\
& 2 & x_1 & + & & x_2 & + & 3 & x_3 & \leq & 7 \\
& & x_1, & & & x_2, & & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



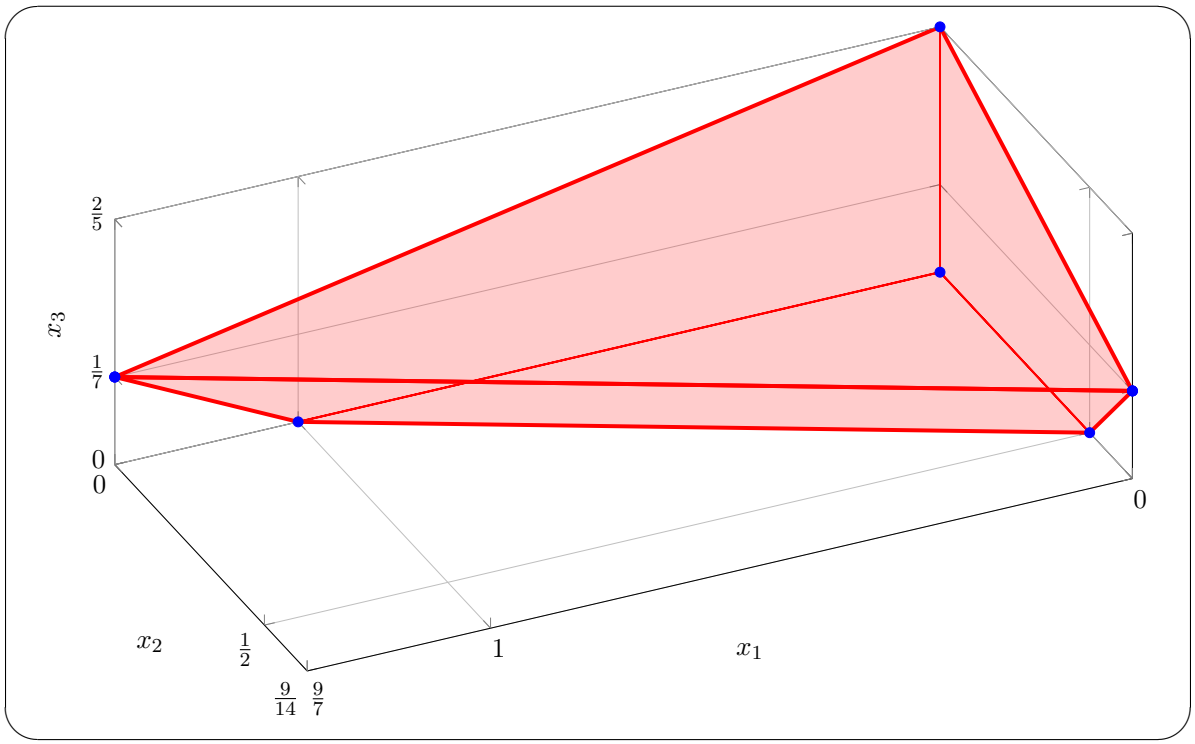
4.3 Problema di PLI *P43*

$$
\begin{array}{lrcrcrcrl}
\max & x_1 & + & 2 & x_2 & + & 7 & x_3 & \\
\text{soggetto a} & x_1 & & & & + & 2 & x_3 & \leq 2 \\
& & & & x_2 & + & 2 & x_3 & \leq 2 \\
& x_1, & & & x_2, & & & x_3 & \in \mathbb{Z}_{\geq 0}
\end{array}
$$



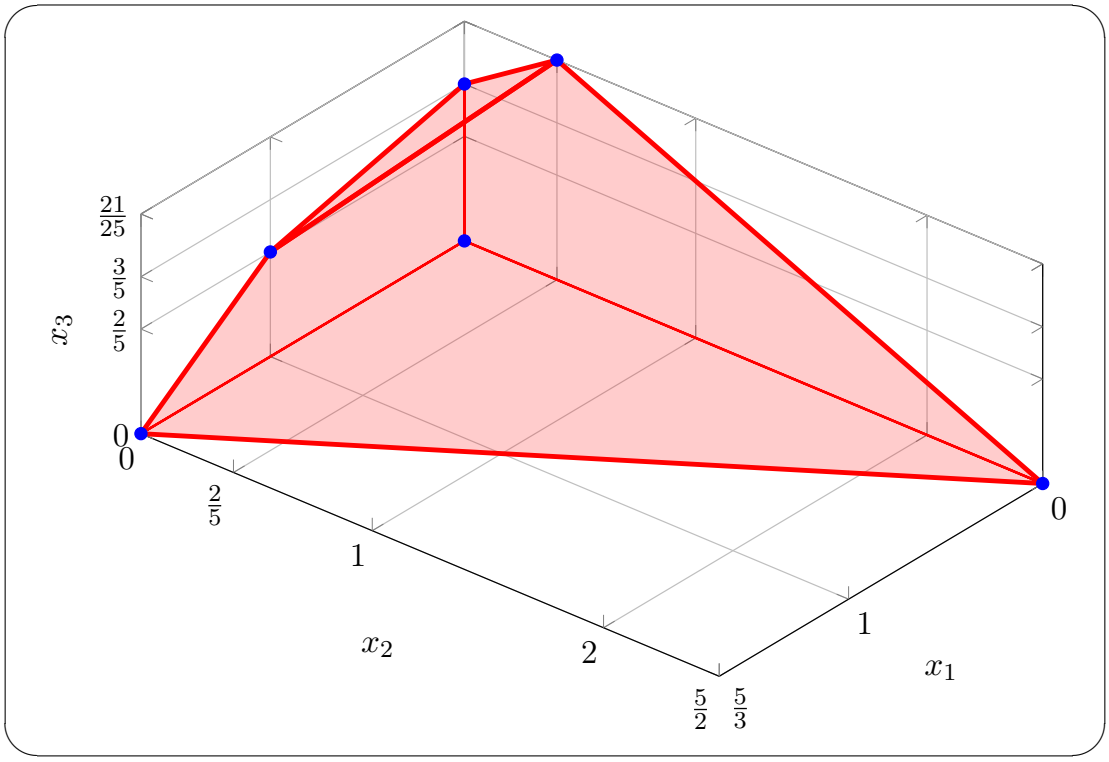
4.4 Problema di PLI *P44*

$$
\begin{array}{lrcrcrcrcl}
\max & x_1 & + & 2\,x_2 & + & 3\,x_3 & & & \\
\text{soggetto a} & x_1 & + & 2\,x_2 & - & 2\,x_3 & \leq & 1 \\
& 2\,x_1 & - & x_2 & - & 3\,x_3 & \leq & 4 \\
& x_1 & + & 2\,x_2 & + & 5\,x_3 & \leq & 2 \\
& x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



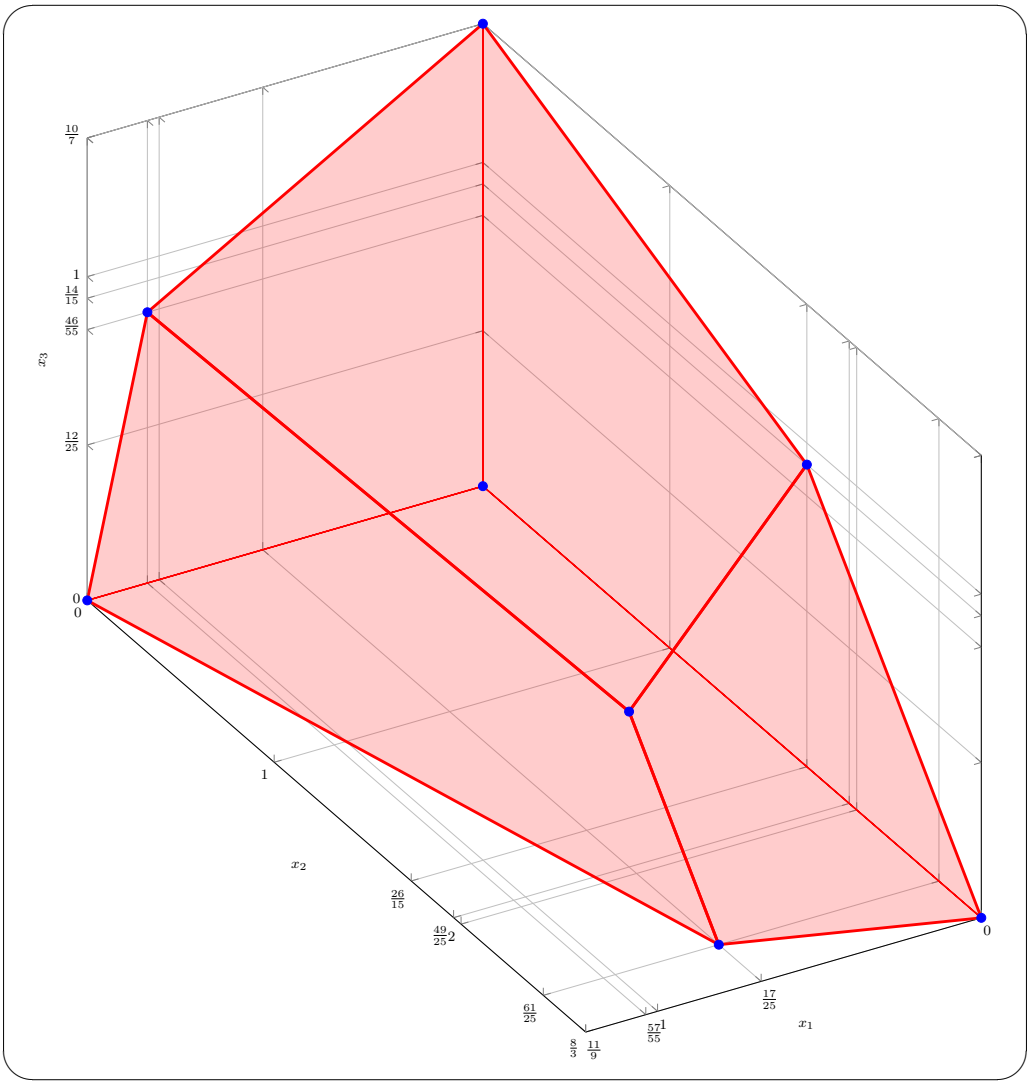
4.5 Problema di PLI *P45*

$$
\begin{array}{rrcrcrcrl}
\max & 4\,x_1 & - & 13\,x_2 & + & 7\,x_3 & & & \\
\text{soggetto a} & 3\,x_1 & + & 2\,x_2 & + & 5\,x_3 & \leq & 5 \\
& x_1 & - & 3\,x_2 & + & 5\,x_3 & \leq & 3 \\
& x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



4.6 Problema di PLI *P46*

$$
\begin{array}{rrcrcrcrl}
\max & 5\ x_1 & + & 6\ x_2 & + & 10\ x_3 & & & \\
\text{soggetto a} & 9\ x_1 & + & 2\ x_2 & + & 2\ x_3 & \leq & 11 \\
& 2\ x_1 & + & 6\ x_2 & + & 6\ x_3 & \leq & 16 \\
& 4\ x_1 & + & 2\ x_2 & + & 7\ x_3 & \leq & 10 \\
& x_1, & & x_2, & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

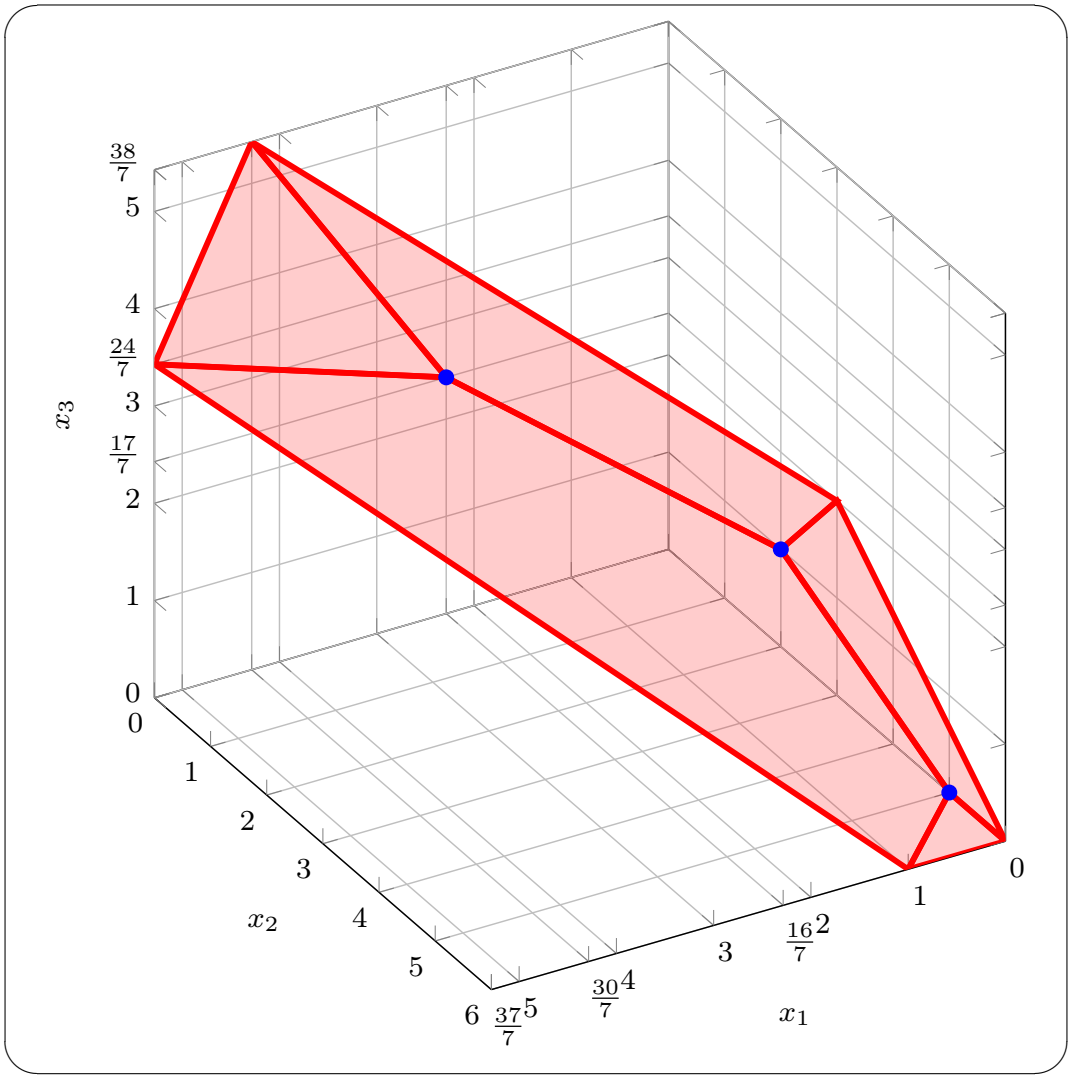


Capitolo 5

Problemi di PLI con tre variabili – Tipo B

5.1 Problema di PLI *P51*

$$
\begin{array}{lrrrrrrrrrl}
\max & - & 2\,x_1 & - & x_2 & & & & & & \\
\text{soggetto a} & & x_1 & - & x_2 & - & 3 & x_3 & \leq & -5 \\
& - & 3 & x_1 & - & 2 & x_2 & + & 2 & x_3 & \leq & -2 \\
& & x_1, & & & x_2, & & & x_3 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



Capitolo 6

Problemi di PLI con tre variabili – Tipo C

6.1 Problema di PLI $P61$

Capitolo 7

Problemi di PLI con più variabili

7.1 Problema di PLI *P71*

$$
\begin{array}{rrrrrrrrrrrrr}
\max & 4 & x_1 & + & & x_2 & + & 5 & x_3 & + & 3 & x_4 & \\
\text{soggetto a} & & x_1 & - & & x_2 & - & & x_3 & + & 3 & x_4 & \leq & 1 \\
& 5 & x_1 & + & & x_2 & + & 3 & x_3 & + & 8 & x_4 & \leq & 55 \\
& - & x_1 & + & 2 & x_2 & + & 3 & x_3 & - & 5 & x_4 & \leq & 3 \\
& & x_1, & & & x_2, & & & x_3, & & & x_4 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$

7.2 Problema di PLI *P72*

$$
\begin{array}{rrrrrrrrrrrrrrrr}
\max & 7 & x_1 & + & 6 & x_2 & + & 5 & x_3 & - & 2 & x_4 & + & 3 & x_5 & \\
\text{soggetto a} & & x_1 & + & 3 & x_2 & + & 5 & x_3 & - & 2 & x_4 & + & 2 & x_5 & \leq & 4 \\
& 4 & x_1 & + & 2 & x_2 & - & 2 & x_3 & + & & x_4 & + & & x_5 & \leq & 3 \\
& 2 & x_1 & + & 4 & x_2 & + & 4 & x_3 & - & 2 & x_4 & + & 5 & x_5 & \leq & 5 \\
& 3 & x_1 & + & & x_2 & + & 2 & x_3 & - & & x_4 & - & 2 & x_5 & \leq & 1 \\
& & x_1, & & & x_2, & & & x_3, & & & x_4, & & & x_5 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$